

Setup

The following code sets up the system by downloading the library from the web.

Make a closed figure function

Here is a list of colors: "blue", "green", "red", "cyan", "magenta", "yellow", "black", "white", "olive", "indigo".

Make a video of drawing a face

```
In [18]: # Playing with b

# Create a graph object with equal_axes)
plot = graph_funs(equal_axes=True, hide_axes=False)

# Setup the grid
# Determine the range for X based on reflections and translations.
# Determine the range for Y based on reflections and translations (look at maximum values).
plot.set_XY(minX=-10, maxX=+10, minY=0, maxY=+30)

# Draw different parts of the face and save in different frames

# Frame 1: draw the face:
make_closed_fig(plot, a=1, b=1.5, h=0, k=0, d=6.5, num_of_points=20)
plot.plot_funs(plot_title="Drawing the face", img_name="frame1.png")

# Frame 2: Draw the eyes:
make_closed_fig(plot, a=0.5, b=2, h=-3, k=12, d=2, num_of_points=20)
make_closed_fig(plot, a=0.5, b=2, h=3, k=12, d=2, num_of_points=20)
plot.plot_funs(plot_title="Drawing the eyes", img_name="frame2.png")

# Frame 3: Draw the mouth:
make_closed_fig(plot, a=1, b=1, h=0, k=6, d=1, num_of_points=20)
plot.plot_funs(plot_title="Drawing the mouth", img_name="frame3.png")

# Frame 3: Draw the nose:
make_closed_fig(plot, a=1, b=1.5, h=0, k=8, d=1.5, num_of_points=20)
plot.plot_funs(plot_title="Drawing the nose", img_name="frame4.png")

# Frame 2: Draw the ear:
make_closed_fig(plot, a=1.2, b=2, h=-8, k=8, d=2, num_of_points=20)
make_closed_fig(plot, a=1.2, b=2, h=8, k=8, d=2, num_of_points=20)
plot.plot_funs(plot_title="Drawing the ear", img_name="frame5.png")

# Frame 2: Draw the eye lids:
make_closed_fig(plot, a=0.5, b=1, h=-3, k=13.80, d=2, num_of_points=20)
make_closed_fig(plot, a=0.5, b=1, h=3, k=13.80, d=2, num_of_points=20)
plot.plot_funs(plot_title="Drawing the eye lids", img_name="frame6.png")

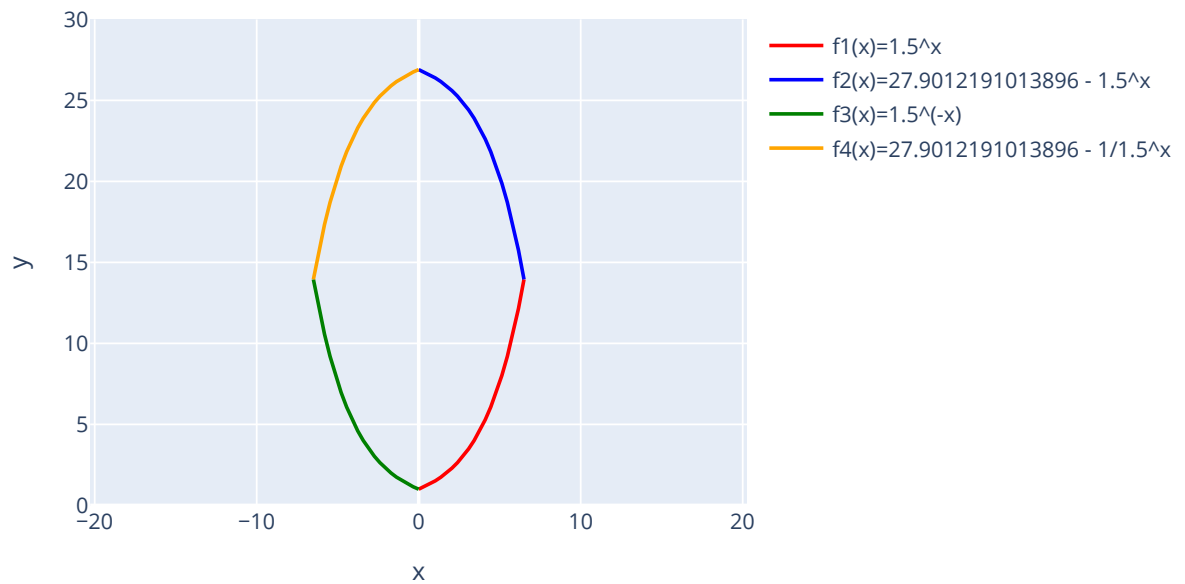
# Frame 2: Draw the brows:
make_closed_fig(plot, a=0.5, b=1, h=-3, k=16, d=2, num_of_points=20)
```

```
make_closed_fig(plot, a=0.5, b=1, h=3, k=16, d=2, num_of_points=20)
plot.plot_funs(plot_title="Drawing the brows", img_name="frame7.png")

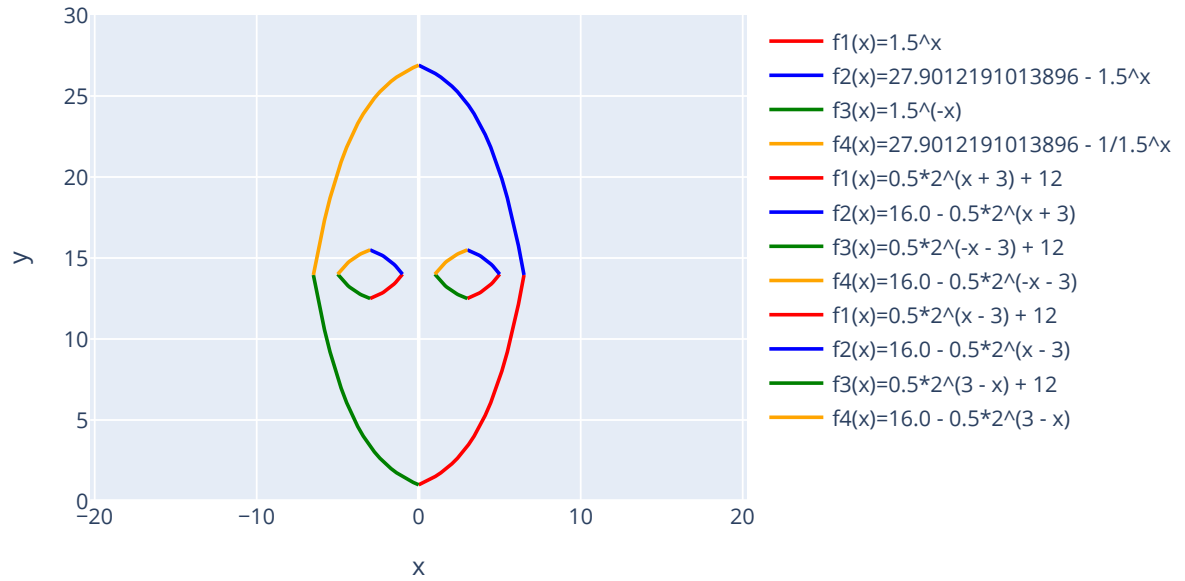
# Frame 2: Draw the er:
make_closed_fig(plot, a=1.2, b=1.5, h=-8, k=7.5, d=1.2, num_of_points=20)
make_closed_fig(plot, a=1.2, b=1.5, h=8, k=7.5, d=1.2, num_of_points=20)
plot.plot_funs(plot_title="Drawing the er", img_name="frame8.png")

# Frame 2: Draw the er:
make_closed_fig(plot, a=3, b=1.5, h=-8, k=8, d=1.2, num_of_points=20)
make_closed_fig(plot, a=3, b=1.5, h=8, k=8, d=1.2, num_of_points=20)
plot.plot_funs(plot_title="Drawing the er", img_name="frame9.png")
```

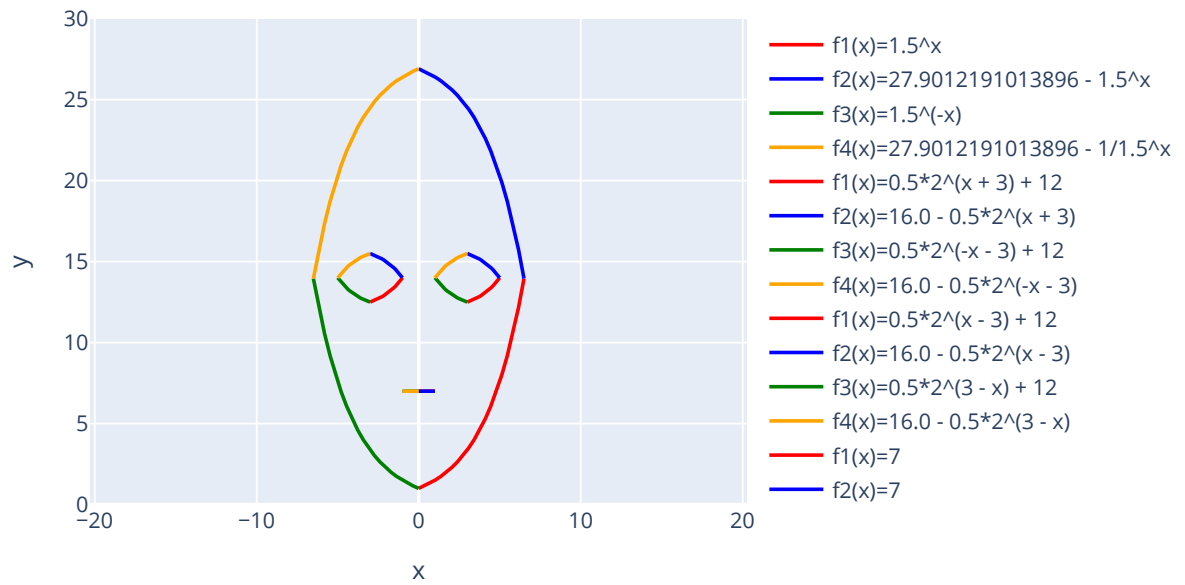
Drawing the face



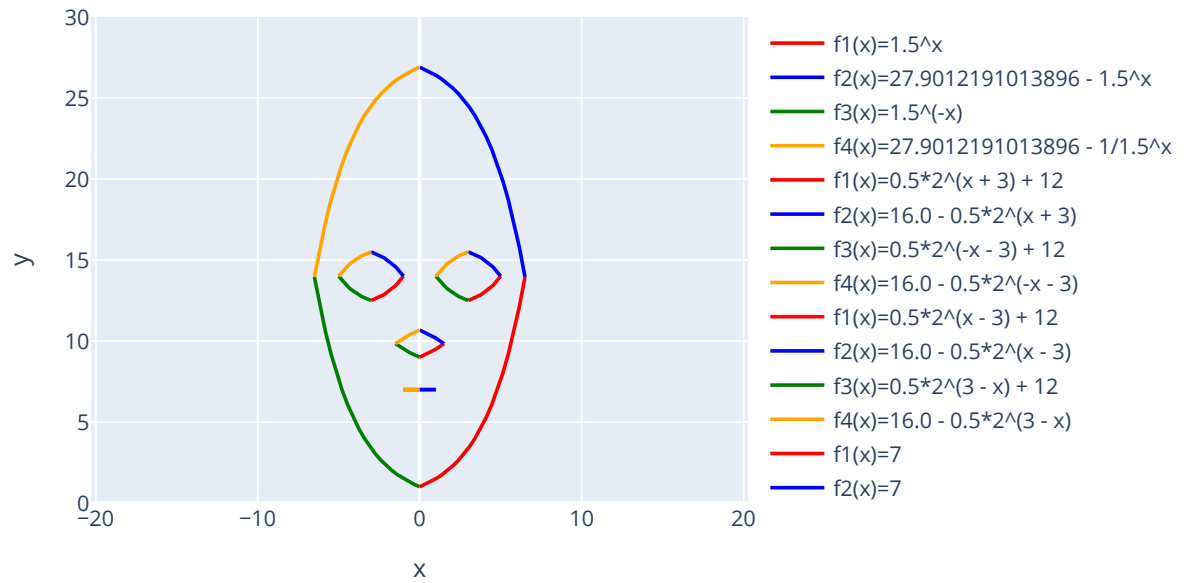
Drawing the eyes



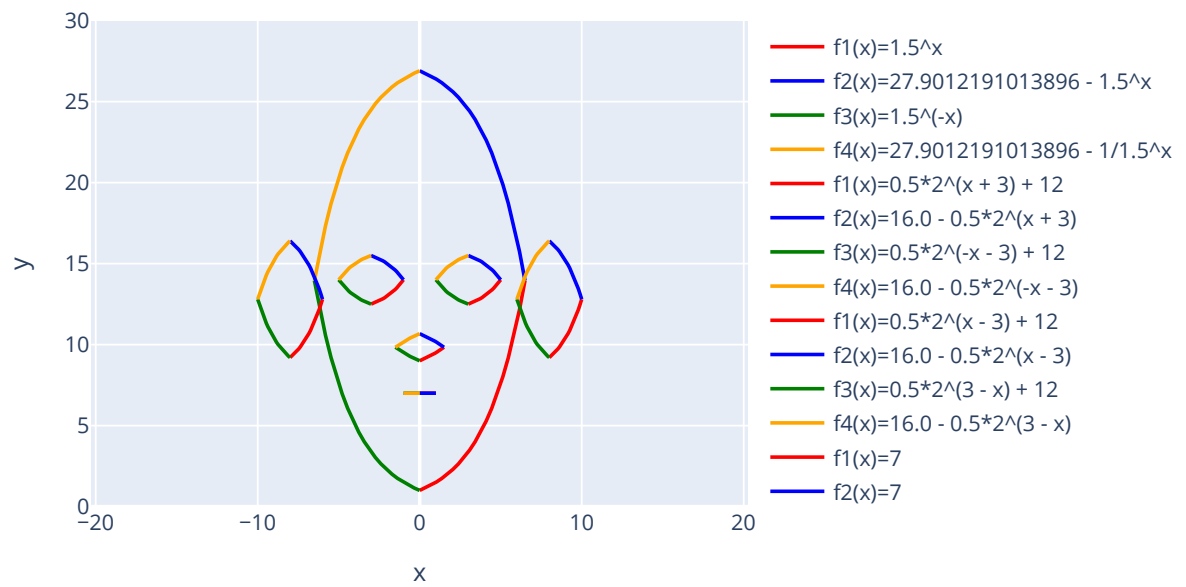
Drawing the mouth



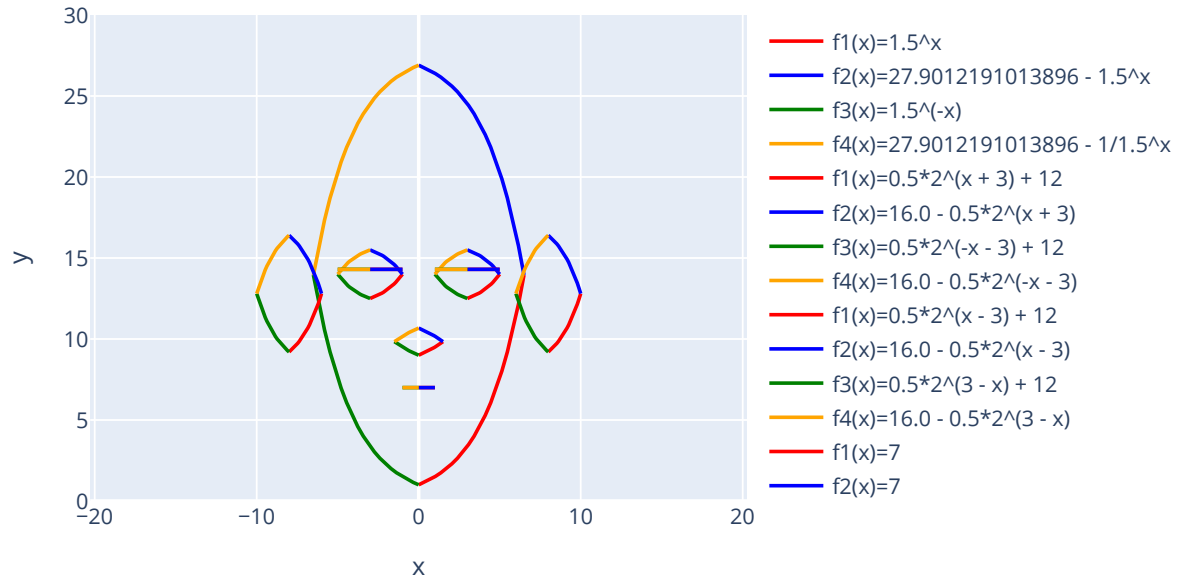
Drawing the nose



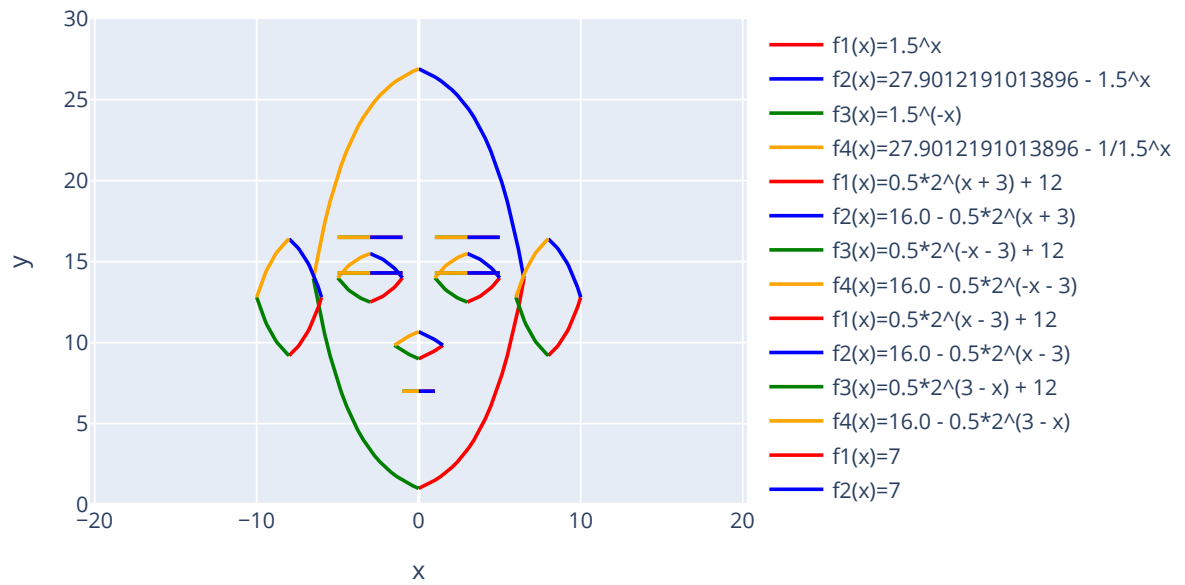
Drawing the er



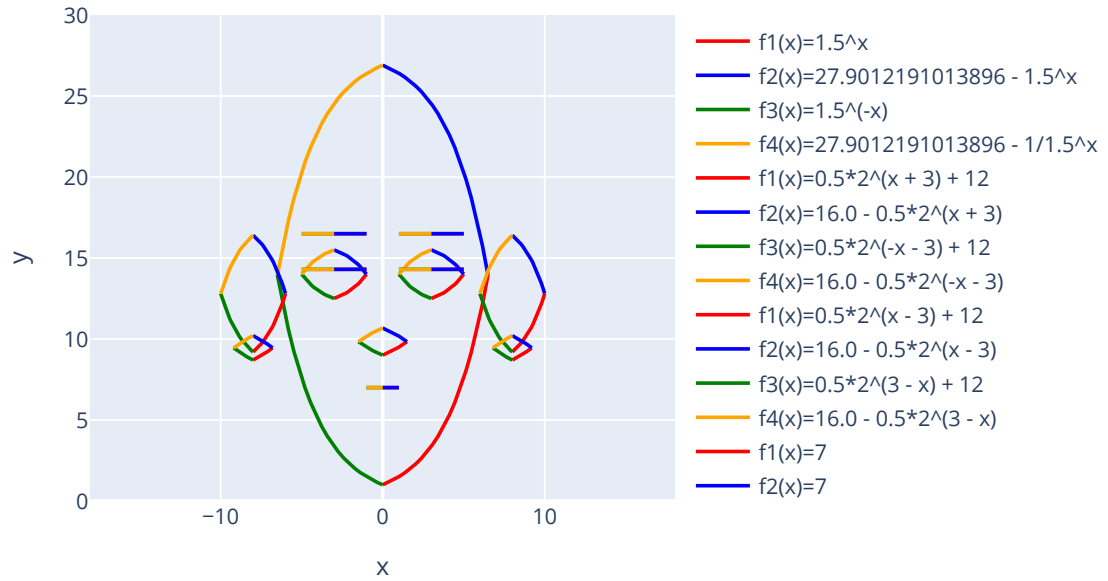
Drawing the eye lids



Drawing the brows



Drawing the er



Drawing the er

